

Survey	Venue	Yr	α mode	β cap.	γ family	δ metric	ϵ catalog
Evaluation of Embodied AI [?]	Authorea	2026	~	~	✗	✗	✓
VLA Datasets & Bench. [?]	arXiv	2026	✗	✗	✗	✗	✓
Benchmark Construction [?]	arXiv	2026	✗	~	✗	✗	✓
RL Reproducibility [?]	CoRL	2020	✗	✗	✗	✗	✗
World Models for Robots [?]	arXiv	2026	~	✗	~	~	~
Embodied World Models [?]	preprint	2026	~	✗	~	✗	~
WM Evaluation (<i>pos.</i>) [?]	arXiv	2026	✓	~	✗	~	✗
Embodied AI: Simulators [?]	TETCI	2022	~	~	✗	✗	✓
Ours (this survey)	--	2026	✓	✓	✓	✓	✓

Table 1. **Topical coverage of the closest benchmark/evaluation surveys versus ours.** ✓ covered as an organising axis, ~ partial or mentioned in passing, ✗ absent. Axes: α evaluation mode (open- vs closed-loop as the *primary* axis); β capability cuts (hidden-state, counterfactual, long-horizon, social); γ capability $\times$ model-family cross-tab (direct VLA vs latent / action-conditioned / generative world models under one protocol); δ behavioural advantage-of-prediction metric (*assessed, not adopted*; e.g. ACTION-ATLAS’s World Advantage Score); ϵ comprehensive benchmark catalogue. This survey is the only work that covers γ , and δ is otherwise reached only *partially*, and only by a *position* paper [?], never by a survey. Venues: CoRL, IEEE TETCI; the remaining rows are arXiv or unrefereed preprints (Authorea, Preprints.org).

Table 2. **Detailed comparison of a representative 31 of the 159 benchmarks** in the landscape (Table ??), grouped by evaluation lane. **Eval:** open open-loop prediction/generation quality, closed closed-loop task success, bridge prediction-to-action bridge. **VLA vs WM:** ✓ native direct-policy vs world-model contrast, ~ partial, -- model-agnostic. Venues are web-verified; “arXiv” marks preprint-only works.

Benchmark	Year	Venue	Eval	Capability	VLA vs WM
§3.1 Policy / manipulation suites					closed
CALVIN [?]	2021	RA-L	closed	long-horizon language	--
LIBERO [?]	2023	NeurIPS D&B	closed	short-horizon + transfer	--
Meta-World [?]	2020	CoRL	closed	multi-task / meta-RL	--
ManiSkill2 [?]	2023	ICLR	closed	generalizable skills	--
SIMPLER [?]	2024	CoRL	closed	generalization (real vs sim)	--
THE COLOSSEUM [?]	2024	RSS	closed	generalization / robustness	--
RoboCasa [?]	2024	RSS	closed	long-horizon (data scaling)	--
RoboArena [?]	2025	CoRL	closed	real-world generalization	--
GemBench [?]	2024	ICRA	closed	generalization levels	--
VLABench [?]	2025	ICCV	closed	long-horizon reasoning	--
§3.2 Embodied navigation, planning & social					closed
Habitat 3.0 [?]	2023	ICLR	closed	social / multi-agent	--
BEHAVIOR-1K [?]	2024	CoRL	closed	long-horizon household	--
ALFRED [?]	2019	CVPR	closed	long-horizon, partial-obs	--
TEACH [?]	2021	AAAI	closed	hidden-state (dialog)	--
RoboTHOR [?]	2020	CVPR	closed	navigation / sim-to-real	--
SocNavBench [?]	2021	ACM THRI	closed	social navigation	--
LIBERO-Mem [?]	2025	AAAI	closed	hidden-state / occlusion	✓

Construct		Counts (a benchmark has it if and only if...)	Does not count
Open-loop (open)	eval	it scores prediction or generation <i>quality</i> on fixed inputs without executing the prediction in a feedback loop: video fidelity (FVD), physics or rollout consistency, or frame-level accuracy	any protocol where an action is executed and the environment responds; offline scoring of a policy’s success
Closed-loop (closed)	eval	success is measured by running a policy <i>in</i> the environment with state feedback and scoring task outcome: success rate, sub-goal completion, or reward under interaction	scoring a generated video or a static prediction; a metric computed without stepping the environment
Bridge (bridge)		a prediction (imagined rollout, video, or latent plan) is <i>converted into executed actions</i> and scored on the resulting task success	a benchmark that stops at prediction quality; one that never turns a prediction into a control signal
Native VLA-vs-WM contrast (γ)		the benchmark <i>itself</i> runs both a direct Vision-Language-Action policy and a world-model policy under one protocol and reports the head-to-head	a model-agnostic suite that merely <i>hosts</i> either family; a suite evaluated with only one family in the original paper
Advantage-of-prediction (δ)		a metric quantifies the closed-loop <i>gain</i> of a predictive policy over a matched direct-VLA baseline on the same tasks (assessed, not adopted)	absolute success with no matched baseline; open-loop quality scores; a ranking that never isolates the predictive component

Table 3. Operational definitions used to place each benchmark on the evaluation axes (§??). They resolve the ambiguous boundary cases: hosting a policy is not the same as *building* a VLA-vs-world-model contrast (γ), and an absolute success number is not an advantage-of-prediction measurement (δ) without a matched baseline. **open** open-loop, **closed** closed-loop, **bridge** bridge.

Table 2 (continued)

Benchmark	Year	Venue	Eval	Capability	VLA vs WM
VIMA-Bench [?]]	2023	ICML	closed	multimodal-prompt generalization	--
§3.3 World-model & video-generation evaluation					open
WorldModelBench [?]]	2025	arXiv	open	video-WM prediction quality	--
Physics-IQ [?]]	2025	WACV	open	physical realism	--
VBench-2.0 [?]]	2025	arXiv	open	generation faithfulness	--
WorldScore [?]]	2025	ICCV	open	world-gen quality	--
EWMBench [?]]	2025	arXiv	open	scene/motion/semantic quality	--
EVA-Bench [?]]	2024	ICML	open	embodied video anticipation	--
WorldPrediction [?]]	2025	arXiv	open	counterfactual action recognition	--
Physion [?]]	2021	NeurIPS D&B	open	physical prediction from vision	--
Physion++ [?]]	2023	NeurIPS D&B	open	physical prediction with online property inference	--
§3.4 Bridges (prediction to action)					bridge
WorldSimBench [?]]	2024	ICML	bridge	video-to-action consistency	~
World-in-World [?]]	2025	ICLR	bridge	closed-loop task success of WMs	~
RoboWM-Bench [?]]	2026	arXiv	bridge	prediction executability	~
WorldArena [?]]	2026	arXiv	bridge	closed-loop WM arena	~

Dimension	LIBERO [?]	Habitat 3.0 [?]	WorldModel-Bench [?]	World-in-World [?]
	<i>policy suite</i>	<i>embodied nav</i>	<i>world-model eval</i>	<i>bridge</i>
What it scores	task success	social-nav success	video-pred. quality	closed-loop WM success
Evaluation mode	closed	closed	open	bridge
Executes actions in a loop?	✓	✓	✗	✓
Native VLA-vs-WM contrast?	✗	✗	✗	~
Capability stressed	short-horizon + transfer	social / multi-agent	physical realism	plan executability
Per-capability slicing?	~	~	✗	✗
Advantage-of-prediction metric?	✗	✗	✗	~
Real / sim / video	sim	sim	video	sim
Counterfactual probing?	✗	✗	✗	✗
Reusable leaderboard?	✓	✓	✓	~

Table 4. Capability matrix, one representative benchmark per evaluation lane. No lane’s exemplar builds a native VLA-vs-world-model contrast: policy and embodied suites are model-agnostic (they *host* any policy), world-model suites score generation quality with no action, and the bridge exemplar reaches closed-loop success but reports only a partial contrast (“visual quality is not task success”). Counterfactual probing is absent throughout. ✓ yes, ~ partial, ✗ no, -- n/a; *open* open-loop, *closed* closed-loop, *bridge* bridge.

	Policy suites [?]	Embodied nav / social [?]	World-model eval [?]	Bridges [?]
Loop closure	✓ closed-loop; steps the environment [?]	✓ closed-loop; long-horizon rollouts [?]	✗ open-loop; scores generation, no action [?]	~ converts a prediction into an executed action [?]
Task horizon	~ short reactive skills; long by chaining [?]	✓ long-horizon household and dialog [?]	~ short clips; temporal but not task-level [?]	~ a single executed rollout [?]
Capability breadth	~ manipulation + transfer; hidden-state rare [?]	✓ navigation, social, some hidden-state [?]	~ physical / temporal; counterfactual barely [?]	✗ narrow; executability only [?]
Model-family contrast	✗ model-agnostic host of any policy [?]	~ memory-vs-Markovian in one suite [?]	✗ world-model-only scores [?]	~ WM as policy substrate; no VLA pole [?]
Reproducibility	~ sim; some real-vs-sim transfer [?]	~ sim-heavy; a sim-to-real gap [?]	✓ fixed offline sets; deterministic scoring [?]	✗ new, still-unstandardised protocols [?]
Structural blind spot	hosts a world-model policy but never builds the VLA contrast	capability cuts exist, but no per-capability family ablation	visual quality is not task success; no behavioural utility	reaches action, but omits the per-capability VLA head-to-head

Table 5. **Evaluation-lane comparison matrix:** the four benchmark lanes on six evaluation axes (loop closure, task horizon, capability breadth, model-family contrast, reproducibility, and characteristic blind spot). Marks grade each lane on the axis (✓ favourable, ~ mixed, ✗ weak), with an anchoring benchmark per cell. No lane closes the loop, slices capability, *and* builds a native VLA-vs-world-model contrast at once; the bridge lane comes closest yet still omits the per-capability VLA head-to-head, which is the empty cell this survey names (§??).

Table 7. **Capability deep-dive:** the benchmarks that probe each of the β capability cuts, and whether any builds a native VLA-vs-world-model contrast. Hidden-state and counterfactual reasoning are touched by only a handful of benchmarks, and only LIBERO-Mem builds a contrast; every other cut is measured model-agnostically. **Eval:** *open/closed*. **VLA vs WM:** ✓ native contrast, ~ partial, -- model-agnostic.

Benchmark	Year	Eval	What it probes	VLA vs WM
Hidden-state / memory / occlusion				

Table 7 (continued)

Benchmark	Year	Eval	What it probes	VLA vs WM
LIBERO-Mem [?]	2025	closed	occluded object state carried across time	✓
TEACH [?]	2021	closed	partner belief and history via dialog	--
Counterfactual / physical inference				
WorldPrediction [?]	2025	open	discriminate correct vs counterfactual action	--
Physion [?]	2021	open	predict a physical outcome from vision	--
Physion++ [?]	2023	open	infer latent physical properties online	--
Long-horizon / compositional				
CALVIN [?]	2021	closed	chain several language subtasks in a row	--
BEHAVIOR-1K [?]	2024	closed	1000 long-horizon household activities	--
ALFRED [?]	2019	closed	instruction to action under partial view	~
Social / multi-agent				
Habitat 3.0 [?]	2023	closed	human-robot co-habitation and hand-off	--
SocNavBench [?]	2021	closed	socially compliant navigation	--

Metric	Definition	Instantiation (testbed)	Axis stressed
Advantage-of-prediction curve	closed-loop success <i>gain</i> of a predictive policy over a matched direct-VLA baseline, reported per capability as a curve, not a single endpoint	a capability-sliced protocol over LIBERO [?] and CALVIN [?] that runs a world-model policy and a VLA policy under one harness	γ family, δ metric
Counterfactual accuracy	task success on episodes that require inference over unobserved or hypothetical dynamics (occlusion, object permanence, “what if”)	WorldPrediction [?] plus occlusion / memory splits in the style of LIBERO-Mem [?]	β capability
Prediction-to-action fidelity gap	the difference between a world model’s open-loop generation quality and its closed-loop task success (“visual quality is not task success”)	bridge protocols such as WorldSimBench [?] and World-in-World [?]	α mode
Per-capability contrast coverage	the fraction of capabilities for which a benchmark actually runs a VLA-vs-world-model head-to-head under a single protocol	audits of the landscape (Table ??) against a declared contrast schema	γ family, ε catalog

Table 6. **An actionable protocol for the future agenda:** four measurable quantities, their definitions, and concrete testbeds. The first anchors the advantage-of-prediction question (δ) on LIBERO / CALVIN-style suites; the second targets the near-unmeasured counterfactual capability; the third quantifies the prediction-to-action gap the bridges expose; the fourth audits per-capability contrast coverage across the landscape. Each is reported as a distribution or curve, never a single headline number.

Sense of “world model”	What it is	Examples	Act. cond.	Loop	Metric	VLA-cmp.
<i>As a representation (what the model is)</i>						
latent WM (LWM)	latent-dynamics predictor $z_{t+1} = f(z_t, a_t)$	DreamerV3, TD-MPC2	✓	✓	~	~
action-cond. WM (WAM)	action-conditioned future-frame predictor	iVideoGPT, GR-2	✓	~	~	~
generative WM (WFM)	text / image-to-video generator	Sora, Cosmos, Genie	~	✗	✓	✗
<i>As a role (what the model is used for)</i>						
neural simulator	learned environment for policy training / eval	UniSim	✓	✓	~	~
evaluation target	benchmarks that score world models	WorldSimBench, WorldModelBench, EWMBench	--	~	✓	✗

Table 8. The term “world model” spans two orthogonal axes the literature routinely conflates: a *representation* axis (what the model is: latent-state dynamics, action-conditioned video, or generative video, a soft boundary since action-conditioned generators straddle WAM and WFM) and a *role* axis (what it is used for: a neural simulator, or the object a benchmark scores). Columns record whether a sense is **action-conditioned**, is used in a closed **loop**, carries a standard evaluation **metric**, and is readily **VLA-comparable** under one protocol (✓ yes, ~ partial, ✗ no, -- n/a). No single sense is action-conditioned, closed-loop, and VLA-comparable at once, which is why a benchmark that contrasts a world-model policy with a VLA policy per capability (§??) is still missing. Evaluation benchmarks score the world model as an *object*; they are not world models acting as judges.